

PROBLEM-4

Given:

Confusion matrices for h_1, h_2, h_3

We will compute:

TPR (True Positive Rate)

$$TPR = \frac{TP}{TP + FN}$$



FPR (False Positive Rate):
$$FPR = \frac{FP}{FP + TN}$$

Distance from perfect classifier
$$D = \sqrt{(FPR - 0)^2 + (1 - TPR)^2}$$

→ For h_1

$$TP = 29, FN = 2$$

$$FP = 1, TN = 13$$

$$TPR = \frac{29}{31} = 0.935$$

$$FPR = \frac{1}{14} = 0.071$$

$$D_1 = \sqrt{(0.071)^2 + (1 - 0.935)^2} = \sqrt{0.0092} \approx 0.096$$

→ For h_2

$$TP = 29, FN = 1$$

$$FP = 3, TN = 12$$

$$TPR = \frac{29}{30} = 0.967$$

$$FPR = \frac{3}{15} = 0.2$$

$$D_2 = \sqrt{(0.2)^2 + (1 - 0.967)^2} = \sqrt{0.0411} \approx 0.203$$

→ For h_3

$$TP = 27, FN = 3$$

$$FP = 3, TN = 12$$

$$TPR = \frac{27}{30} = 0.9$$

$$FPR = \frac{3}{15} = 0.2$$



$$D_3 = \sqrt{(0.2)^2 + (1-0.9)^2} = \sqrt{0.05} \approx 0.224$$

Hypothesis	TPR	FPR	Distance
h_1	0.935	0.071	0.096
h_2	0.967	0.2	0.203
h_3	0.9	0.2	0.224

h_1 has the smallest distance
But classifier: h_1

→ FP cost = 4 × FN cost

$$D = \sqrt{(4 \cdot \text{FPR})^2 + (1 - \text{TPR})^2}$$

for h_1 :

$$D_1 = \sqrt{(4 \times 0.071)^2 + (0.065)^2} = \sqrt{0.0849} \approx 0.291$$

for h_2

$$D_2 = \sqrt{(4 \times 0.2)^2 + (0.033)^2} = \sqrt{0.6411} \approx 0.801$$

for h_3

$$D_3 = \sqrt{(4 \times 0.2)^2 + (0.1)^2} = \sqrt{0.65} \approx 0.806$$

Hypothesis	Weighted Distance
h_1	0.291
h_2	0.801
h_3	0.806

When FP and FN have equal cost
 h_1 is ~~the~~ best classifier



Date. / /

Page No.

when FP cost is $4 \times FN$

Best classifier is still h_1 .